

Software Requirements Specification

3.1 Functional Requirements

FR-01 — User Registration / Profile

- **FR-01.1** The system shall allow users to create an account with basic profile information.
- **FR-01.2** The system shall capture user preferences including language(s), interests (architecture, religion, photography, history, etc.), and accessibility requirements (mobility, visual, auditory).
- **FR-01.3** The system shall allow users to set preferred tour depth (brief, standard, deep-dive) and prior knowledge level.
- **FR-01.4** The system shall support guest/anonymous mode with limited personalization.

FR-02 — Heritage Site Discovery

- **FR-02.1** The system shall provide search functionality by site name, location, period, style, and UNESCO status.
- **FR-02.2** The system shall categorize sites by type (temple, fort, mosque, cave, palace, stepwell, etc.), era, and significance.
- **FR-02.3** The system shall display nearby heritage sites based on user GPS location.
- **FR-02.4** The system shall recommend lesser-known sites based on user interests, itinerary, and proximity.

FR-03 — AI Journey Planner

- **FR-03.1** The system shall generate personalized itineraries based on available time (30 min to full day).
- **FR-03.2** The system shall factor in user budget constraints for entry fees, transport, and experiences.
- **FR-03.3** The system shall adapt routes to user interests, mobility needs, and prior knowledge.
- **FR-03.4** The system shall integrate real-time weather data to suggest optimal visit times and shaded routes.
- **FR-03.5** The system shall incorporate traffic and transport information for last-mile planning.
- **FR-03.6** The system shall use crowd condition data to recommend less congested time slots and routes.

FR-04 — Verified Heritage Content

- **FR-04.1** The system shall deliver historical information including timelines, phases, and key events for each site.
- **FR-04.2** The system shall provide architectural analysis including style, materials, and structural features.
- **FR-04.3** The system shall include cultural significance, community narratives, and intangible heritage context.
- **FR-04.4** The system shall display source citations for all factual claims (ASI, UNESCO, expert-reviewed).
- **FR-04.5** The system shall label each feature as original, restored, reconstructed, or interpretive.
- **FR-04.6** The system shall distinguish between archaeological evidence, local tradition, and scholarly interpretation.

FR-05 — Location-Based Triggering

- **FR-05.1** The system shall use GPS to detect user location and trigger zone-level content (3–10 m accuracy).
- **FR-05.2** The system shall use geofencing to activate structure-level content within defined boundaries.
- **FR-05.3** The system shall support QR code scanning for precise point-of-interest information.
- **FR-05.4** The system shall allow manual selection of structures on an interactive map when GPS is unreliable.
- **FR-05.5** The system shall optionally use AR markers for visual-inertial positioning on compatible devices.

FR-06 — Navigation

- **FR-06.1** The system shall guide users to site entrances from their current location.
- **FR-06.2** The system shall provide internal site routes connecting key structures, viewpoints, and galleries.
- **FR-06.3** The system shall navigate users to facilities including toilets, drinking water, first aid, and exits.
- **FR-06.4** The system shall offer accessible routes with step-free paths, ramp locations, and battery-vehicle points.
- **FR-06.5** The system shall highlight recommended viewpoints and photography spots.

FR-07 — Multilingual Interaction

- **FR-07.1** The system shall deliver content in Hindi and major Indian languages (Tamil, Telugu, Kannada, Malayalam, Bengali, Marathi, Gujarati, Punjabi, Odia, Urdu, etc.).
- **FR-07.2** The system shall support key foreign languages (English, French, Spanish, German, Japanese, Chinese, Arabic, etc.).
- **FR-07.3** The system shall provide text and audio output in the selected language.
- **FR-07.4** The system shall support speech-to-text input for user queries.
- **FR-07.5** The system shall support text-to-speech output for all textual content.
- **FR-07.6** The system shall provide real-time translation for chat-based interactions.

FR-08 — Offline Mode

- **FR-08.1** The system shall allow users to download complete site packs over Wi-Fi before visiting.
- **FR-08.2** Offline site packs shall include vector maps, audio tours, full text content, and facility information.
- **FR-08.3** The system shall cache QR-linked content for offline access.
- **FR-08.4** The system shall include selected AR/3D assets in offline packs with LOD optimization.
- **FR-08.5** The system shall sync updates and usage data when connectivity is restored.

FR-09 — AR / 3D

- **FR-09.1** The system shall display historical reconstruction overlays showing structures in their original state.
- **FR-09.2** The system shall support object/structure recognition via camera for AR content triggering.
- **FR-09.3** The system shall provide interactive 3D overlays with architectural annotations.
- **FR-09.4** The system shall gracefully degrade to 2D images, text, and audio on non-AR-capable devices.

FR-10 — Guide Verification

- **FR-10.1** The system shall maintain profiles for licensed local guides with credential information.
- **FR-10.2** The system shall support QR-based authentication for guide identity and license status.

- **FR-10.3** The system shall display guide specialization, languages spoken, and user ratings.

FR-11 — Local Marketplace

- **FR-11.1** The system shall list verified local guides available for hire at each site.
- **FR-11.2** The system shall showcase artisan profiles, crafts, and purchase options.
- **FR-11.3** The system shall list homestays and local accommodation near heritage clusters.
- **FR-11.4** The system shall offer cultural experiences, workshops, and community-led walks.

FR-12 — Visitor Management

- **FR-12.1** The system shall display real-time or estimated crowd levels at sites and zones.
- **FR-12.2** The system shall notify users of temporary closures, restricted areas, and conservation work.
- **FR-12.3** The system shall issue conservation alerts for sensitive zones (e.g., no-flash areas, restricted structures).
- **FR-12.4** The system shall provide aggregated, anonymized visitor analytics to site managers via an admin dashboard.

FR-13 — Admin / CMS

- **FR-13.1** The system shall provide a content management system for ASI/state editors and expert reviewers.
- **FR-13.2** The CMS shall support expert verification workflows with approval stages.
- **FR-13.3** The CMS shall maintain version control with change logs, dates, and author metadata.
- **FR-13.4** The system shall allow real-time updates for opening hours, ticket prices, closures, and facility changes.
- **FR-13.5** The system shall provide analytics dashboards for usage patterns, popular content, and visitor flow.

3.2 Non-Functional Requirements

Category	Requirement
Performance	Content triggering within 2 seconds of entering a geofence; audio playback start within 1 second; map rendering within 3 seconds
Scalability	Architecture shall support scaling from pilot (5–10 monuments, 10–50K MAU) to national scale (50+ monuments, millions of users)

Category	Requirement
Availability	99.5% uptime for cloud services; 100% availability for downloaded offline content
Security	End-to-end encryption for user data; secure authentication; API key management; OWASP compliance
Privacy	GDPR/DPDP Act compliance; anonymized analytics; explicit consent for location tracking; data minimization
Offline Reliability	Full core functionality (maps, audio, text, QR content) available without internet; graceful sync on reconnection
Accessibility	WCAG 2.1 AA compliance; screen-reader compatibility; high-contrast mode; adjustable font sizes; accessible route information
Multilingual Accuracy	Human-reviewed translations for core heritage content; machine translation with confidence indicators for dynamic content
Low-End Device Compatibility	Basic mode (text + audio + 2D maps) functional on devices with 2 GB RAM and Android 10; AR features optional
Battery Efficiency	GPS polling optimized to minimize drain; AR/camera usage clearly indicated with battery warnings; low-power mode available
Maintainability	Modular architecture; CI/CD pipelines; documented APIs; content separated from code
Data Integrity	Source provenance for all heritage content; version control; audit trails for content changes
Content Provenance	Every piece of heritage information traceable to its source (ASI, UNESCO, expert, community); clear labeling of AI-generated explanations

3.3 Hardware Requirements

Minimum Smartphone (Basic Mode):

- Android 10+ or iOS 14+
- GPS capability
- Camera (for QR scanning)
- Internet capability (for initial download)
- 2 GB RAM
- 500 MB free storage per site pack

Recommended Smartphone (Full AR Mode):

- Android 12+ with ARCore support or iOS 15+ with ARKit
- 4 GB+ RAM
- Decent GPU (Snapdragon 7+ Gen 2 / Dimensity 8200+ or equivalent)
- Dual-frequency GPS (L1+L5) for improved accuracy
- 2 GB+ free storage for AR/3D assets

ARCore Coverage: By 2026, 1,200+ Android models support ARCore 1.40+, including Pixel 6+, Samsung S22/S23/S24, OnePlus 10T+, and many mid-tier phones. India's device mix is dominated by mid-tier Android (Vivo ~19–20%, Samsung ~14% by volume), so non-AR fallbacks are essential.

Optional Infrastructure:

- Cloud servers (AWS Mumbai/Hyderabad, Azure Pune, GCP Mumbai)
- CDN for content delivery
- GPS/map service subscriptions
- QR code printing and placement at monuments
- 3D scanning/photogrammetry equipment for content creation

3.4 Software / Technology Stack

Layer	Technology Options
Mobile	Flutter / React Native / Native Android (Kotlin); ARCore for Android AR; ARKit for iOS; Unity/AR Foundation for complex 3D scenes
AI	LLM (GPT-4 / Claude / open-source alternatives); RAG pipeline; Vector database (Pinecone / Weaviate / pgvector); Recommendation engine; Agentic workflow for multi-step planning
Backend	Node.js / Python (FastAPI) / ASP.NET Core; REST APIs; GraphQL where needed
Database	PostgreSQL + PostGIS for geospatial data; Vector database for RAG; Redis for caching
Data Storage	Object storage (S3/Azure Blob/GCS) for media, 3D assets, audio; CDN for distribution
CMS	Custom headless CMS with version control, approval workflows, and multilingual support

External APIs	Maps (Google Maps Platform / Mapbox); Weather (OpenWeatherMap / Google Weather); Translation (Cloud NMT / Sarvam / Bhashini); TTS/STT (Google Cloud Speech / Azure Speech / Bhashini)
3D/AR	glTF / 3D Tiles for model delivery; ARCore / ARKit for tracking; LOD streaming for performance
DevOps	Docker; Kubernetes; CI/CD (GitHub Actions / GitLab CI); Monitoring (Prometheus / Grafana)

3.5 Data Requirements

For **every monument** in the system, the following data must be maintained:

Data Field	Description
Name	Official name + local name(s)
Location	GPS coordinates, address, district, state
Historical timeline	Key periods, dynasties, events
Architecture	Style, materials, structural features, phases
Cultural significance	Religious, social, artistic importance
Images	High-resolution photographs, historical images
Audio	Narrated tours in multiple languages
Sources	Citations to ASI, UNESCO, academic, expert records
Conservation status	Current condition, ongoing work, restrictions
Accessibility	Ramps, step-free routes, accessible toilets, parking
Facilities	Toilets, drinking water, first aid, parking, ticket counters
Opening hours	Regular and seasonal timings
Ticket information	Prices (domestic/foreign), online booking links
Routes	Internal paths, recommended circuits, accessible routes
3D/AR assets	Models, textures, animations for AR overlays

Data Field	Description
Local community information	Artisans, guides, homestays, cultural experiences
Last updated date	Timestamp of most recent content review
Verification status	Approved / pending review / community-contributed

3.6 Governance & Verification Requirements

Core Principle: AI should **not** independently establish historical truth. Saarthi acts as an aggregator and presenter, not the primary owner of historical truth.

Verification Pipeline:

Source (ASI/UNESCO/State/Expert)

- Expert Review
- Approval
- Publication
- Versioning
- Update Cycle

Detailed Governance Model:

1. Data Custodianship:

- Primary custodians: ASI (centrally protected ~3,685 sites), state archaeology departments, Monument Mitras / Adopt a Heritage partners.
- Platform role: Aggregator and presenter; data ingested from authoritative sources.

2. Verification Model:

- Core facts (history, architecture, conservation status) verified by ASI/state experts or empanelled scholars.
- Local narratives and oral histories labeled as such, with contributor information and expert review where possible.
- AI-generated explanations must be source-grounded via RAG over approved records and must display citations and "evidence vs. interpretation" labels.

3. **Content Flow:**

- Approved heritage database → RAG retrieval → AI explanation generation → Source citation attachment → Human review (for flagged content) → Publication.

4. **Update Workflow (CMS):**

- Version-controlled CMS where ASI/state editors can update opening hours, ticket prices, closures, conservation alerts, route changes, facility updates, and historical interpretations.
- Each update carries metadata: source, author, date, approval status.
- Users see "last updated" on each page/audio/AR layer.

5. **New Monument Onboarding:**

- Nomination by ASI circle / state tourism / Monument Mitra.
- Data collection: metadata, photos, narratives, accessibility info.
- Digitization prioritization based on footfall, significance, and readiness.
- Content production: narratives, translations, audio, maps, optional 3D/AR.
- Expert review + ASI/state sign-off before publishing.
- Launch, monitoring, feedback collection, and iteration.